higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## MATHEMATICS N6

## 29 MARCH 2019

**This marking guideline consists of 16 pages.**

**The paper is marked out of 200 and divided by 2 to get a mark out of 100.**

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## QUESTION 1

1.1 $z = \tan(xy)$

$$\frac{\partial z}{\partial x} = \sec^2(xy).y \quad ✓✓$$

$$\frac{\partial^2 z}{\partial x^2} = y.2\sec(xy).\sec(xy)\tan(xy).y \quad ✓✓✓✓$$

(6)

1.2 $r = \sqrt{x^2 + y^2}$ ✓

$$\Delta r = \frac{\partial r}{\partial x}\Delta x + \frac{\partial r}{\partial y}\Delta y \quad ✓$$

$$= \frac{x}{\sqrt{x^2 + y^2}}\Delta x + \frac{y}{\sqrt{x^2 + y^2}}\Delta y \quad ✓✓$$

$$= \frac{20}{\sqrt{20^2 + 10^2}}.2 + \frac{10}{\sqrt{20^2 + 10^2}}(-2) \quad ✓$$

$$= \frac{40 - 20}{\sqrt{20^2 + 10^2}} \quad ✓$$

$$= 0,894\,m$$

(6)
**[12]**

**QUESTION 2**

2.1

$$\int y\,dx$$

$$= \int 1 - \tan^4 3x\,dx$$

$$= \int (1 - \tan^2 3x)\left(1 + \tan^2 3x\right)dx \quad ✓✓$$

$$= \int (1 - \tan^2 3x)\left(\sec^2 3x\right)dx \quad ✓$$

$$= \int (\sec^2 3x - \tan^2 3x \sec^2 3x\,dx \quad ✓✓$$

$$= \frac{1}{3}\tan 3x - \frac{1}{3}\frac{\tan^3 3x}{3} + c \quad ✓✓✓$$

Or using $u = \tan 3x$

**Alternative 1**

$$\int y\,dx$$

$$= \int 1 - \tan^4 3x\,dx$$

$$\int (1 - \tan^2 3x.\tan^2 3x\,dx \quad ✓$$

$$= \int (1 - \tan^2 3x\left(\sec^2 3x - 1\right)dx \quad ✓$$

$$= \int (1 - \tan^2 3x \sec^2 3x + \tan^2 3x\,dx \quad ✓✓$$

$$= x - \frac{1}{3}\frac{\tan^3 3x}{3} + \frac{1}{3}\tan 3x - x + c \quad ✓✓✓$$

$$= \frac{1}{3}\frac{\tan^3 3x}{3} + \frac{1}{3}\tan 3x + c \quad ✓$$

$=$

**Alternative 2**

✓ (8)

✓

2.2 (8)

2.3

$$\int (1 - 2\sin^2 2x)^3 dx \quad ✓✓✓$$

$$= \int (\cos 4x)^3 dx \quad ✓✓$$

$$= \int \cos^2 4x \cos 4x\,dx \quad ✓$$

$$= \int (1 - \sin^2 4x)\cos 4x\,dx \quad ✓ \quad ✓✓$$

$$= \int (\cos 4x - \sin^2 4x \cos 4x\,dx \quad ✓$$

$$= \frac{\sin 4x}{4} - \frac{1}{4}\frac{\sin^3 4x}{3} + c \quad ✓✓✓$$

✓

$$\int \left[1 - 2(1 - \cos^2 2x)\right]^3 dx$$

$$= \int \left[-1 + 2\cos^2 2x)\right]^3 dx$$

$$= \int (\cos 4x)^3 dx$$

Or using $u = \sin 4x$

(8)

2.4

$$\int \frac{\sin 2x}{e^{2x}}dx$$

$$f(x)=e^{-2x} \quad g'(x)=\sin 2x$$

$$=\int e^{-2x}\sin 2x dx \quad \checkmark$$

$$=e^{-2x}.-\frac{\cos 2x}{2}-\int -2e^{-2x}.-\frac{\cos 2x}{2}dx \quad \checkmark\checkmark$$

$$=-\frac{1}{2}e^{-2x}\cos 2x-\int e^{-2x}\cos 2x dx$$

$$=-\frac{1}{2}e^{-2x}\cos 2x-\left[e^{-2x}.\frac{\sin 2x}{2}-\int -2e^{-2x}.\frac{\sin 2x}{2}dx\right] \quad \checkmark\checkmark$$

$$=-\frac{1}{2}e^{-2x}\cos 2x-\frac{1}{2}e^{-2x}.\sin 2x-\int e^{-2x}\sin 2x dx$$

$$2\int e^{-2x}\sin 2x dx=-\frac{1}{2}e^{-2x}\cos 2x-\frac{1}{2}e^{-2x}.\sin 2x \quad \checkmark$$

$$\int e^{-2x}\sin 2x dx=-\frac{1}{4}e^{-2x}\cos 2x-\frac{1}{4}e^{-2x}.\sin 2x+c \quad \checkmark\checkmark$$

**Alternative**

$$f(x)=\sin 2x \qquad g'(x)=e^{-2x}$$

(8)

$$\int e^{-2x}\sin 2x dx \quad \checkmark$$

$$= \sin 2x\frac{e^{-2x}}{-2} - \int 2\cos 2x\frac{e^{-2x}}{-2}dx \quad \checkmark\checkmark$$

$$= -\frac{1}{2}\sin 2x\, e^{-2x} + \int e^{-2x}\cos 2x\, dx$$

$$= -\frac{1}{2}\sin 2x\, e^{-2x} + \left[\cos 2x\frac{e^{-2x}}{-2} - \int -2\sin 2x\frac{e^{-2x}}{-2}dx\right] \quad \checkmark\checkmark$$

$$= -\frac{1}{2}\sin 2x\, e^{-2x} - \frac{1}{2}e^{-2x}\cos 2x - \int e^{-2x}\sin 2x\, dx$$

$$2\int e^{-2x}\sin 2x\, dx = -\frac{1}{2}\sin 2x\, e^{-2x} - \frac{1}{2}e^{-2x}\cos 2x + c \quad \checkmark$$

$$\int e^{-2x}\sin 2x\, dx = -\frac{1}{4}\sin 2x\, e^{-2x} - \frac{1}{4}e^{-2x}\cos 2x + c \quad \checkmark\checkmark$$

2.5

$$\int \frac{x^2-2}{x^4-4}dx$$

$$= \int \frac{x^2-2}{(x^2+2)(x^2-2)}dx \quad \checkmark$$

$$= \int \frac{1}{2+x^2}dx \quad \checkmark$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}} + c \quad \checkmark\checkmark$$

(4)
**[36]**

## QUESTION 3

3.1

(12)

$\int \frac{x^2+x-7}{x^2+x-6}dx$ ✓

$\int \frac{(x^2+x-6)-1}{x^2+x-6}dx$ ✓✓

$\int 1-\frac{1}{x^2+x-6}dx$ ✓ Or using long division

$\frac{1}{x^2+x-6}=\frac{1}{(x+3)(x-2)}=\frac{A}{(x+3)}+\frac{B}{(x-2)}$

$1=A(x-2)+B(x+3)$ ✓

$x=2 \quad B=\frac{1}{5}$ ✓

$x=-3 \quad A=-\frac{1}{5}$ ✓

$1=A(x-2)+B(x+3)$

$1=Ax-2A+Bx+3B$

$A+B=0.....(1) \Rightarrow B=-A$

$-2A+3B=1.....(2)$

$-2A+3(-A)=1$

$-5A=1 \Rightarrow A=-\frac{1}{5} \quad and \quad B=\frac{1}{5}$

$\int 1-\frac{1}{x^2+x-6}dx=\int 1-\left[\frac{-\frac{1}{5}}{(x+3)}+\frac{\frac{1}{5}}{(x-2)}\right]dx$

$=x+\frac{1}{5}\ln\ln(x+3)-\frac{1}{5}\ln(x-2)+c$

$=x+\frac{1}{5}\{(\ln(x+3)-\ln(x-2)\}+c$

$=x+\frac{1}{5}\ln\frac{x+3}{x-2}+c$

3.2 $\int \frac{7x^2-12x+8}{(2x-1)(x^2-2x+2)}dx$

$\frac{7x^2-12x+8}{(2x-1)(x^2-2x+2)}=\frac{A}{2x-1}+\frac{Bx+C}{x^2-2x+2}$ ✓✓

$7x^2-12x+8=A(x^2-2x+2)+(Bx+C)(2x-1)$ ✓

$x=\frac{1}{2} \quad 7(\frac{1}{2})^2-12(\frac{1}{2})+8=A\{(\frac{1}{2})^2-2(\frac{1}{2})+2\} \Rightarrow$ ✓ $A=3$ ✓

$7x^2-12x+8=Ax^2-2Ax+2A+2Bx^2+2Cx-Bx-C$ ✓

$A+2B=7 \qquad 3+2B=7$ $B=2$ ✓

$-2A+2C-B=-12$

$-2(3)+2C-2=-12$ $C=-2$ ✓

(12)

$$\int \frac{7x^2-12x+8}{(2x-1)(x^2-2x+2)}dx$$

$$= \int \frac{3}{2x-1} + \frac{2x-2}{x^2-2x+2}dx \quad ✓$$

$$= \frac{3}{2}\ln(2x-1) + \ln(x^2-2x+2) + c$$

**[24]**

## QUESTION 4

4.1 $$\frac{1}{x}\frac{dy}{dx} - \frac{1}{x^2}y = 3x - \sin x + \cos x$$

$$\frac{dy}{dx} - \frac{1}{x}y = x(3x - \sin x + \cos x) \quad ✓$$

$$e^{\int p dx} = e^{\int -\frac{1}{x}dx} \quad ✓$$

$$= e^{-\ln x} \quad ✓$$

$$= e^{\ln x^{-1}}$$

$$= x^{-1} \quad = \frac{1}{x} \quad ✓$$

$$e^{-\ln x} = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

$$\int Q e^{\int p dx} dx = \int x(3x - \sin x + \cos x)\frac{1}{x}dx \quad ✓$$

$$= \int 3x - \sin x + \cos x)dx \quad ✓$$

$$= \frac{3}{2}x^2 + \cos x + \sin x \quad ✓✓$$

$$\frac{y}{x} = \frac{3}{2}x^2 + \cos x + \sin x + c \quad ✓$$

$$x = 1;\ y = 1 \quad \frac{1}{1} = \frac{3}{2}(1)^2 + \cos(1) + \sin(1) + c \quad ✓$$

$$c = -1,882 \quad ✓$$

$$\frac{3}{2}x^2 + \cos x + \sin x - 1,882 \quad ✓$$

(12)

$$\frac{y}{x}=\frac{3}{2}x^2+cosx+sinx-1.882$$

4.2

$$6\frac{d^2y}{dx^2}-\frac{dy}{dx}-y=x^2$$

$$6r^2-r-1=0$$

$$(3r+1)(2r-1)=0 \quad ✓$$

$$r=-\frac{1}{3} \quad r=\frac{1}{2} \quad ✓$$

$$y_c=Ae^{-\frac{1}{3}x}+Be^{\frac{1}{2}x} \quad ✓$$

$$y=Cx^2+Dx+E \quad ✓$$

$$\frac{dy}{dx}=2Cx+D \qquad \frac{d^2y}{dx^2}=2C \quad ✓$$

$$6(2C)-(2Cx+D)-(Cx^2+Dx+E)=x^2 \quad ✓$$

$$12C-2Cx-2D-Cx^2-Dx-E=x^2$$

$$x^2: -C=1 \quad \therefore C=-1 \quad ✓$$

$$x: -2C-2D=0 \quad \therefore D=2 \quad ✓$$

$$12C-D-E=0 \quad \therefore E=-14 \quad ✓$$

$$y_p=-x^2+2x-14 \quad ✓$$

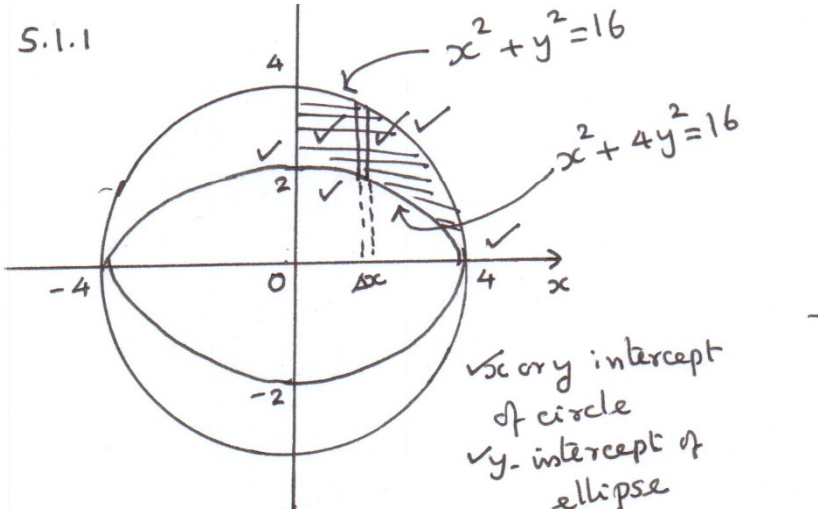
$$y=y_c+y_p$$

$$y=Ae^{-\frac{1}{3}x}+Be^{\frac{1}{2}x}-x^2+2x-14 \quad ✓$$

(12)
**[24]**

## QUESTION 5

5.1 5.1.1




(6)

5.1.2 $Area =$

$$= \int_a^b y_1 - y_2 \, dx \quad ✓$$

$$= \int_0^4 \sqrt{16-x^2} - \frac{1}{2}\sqrt{16-x^2} \, dx \quad ✓✓✓$$

$$= \frac{1}{2}\int_0^4 \sqrt{16-x^2}\, dx \quad ✓$$

$$= \frac{1}{2}\left[\frac{16}{2}\sin^{-1}\frac{x}{4} + \frac{x}{2}\sqrt{16-x^2}\right]_0^4 \quad ✓✓$$

$$= \frac{1}{2}\left[8\sin^{-1}\frac{4}{4} + \frac{4}{2}\sqrt{16-4^2} - \left\{8\sin^{-1}\frac{0}{4} + \frac{0}{2}\sqrt{16-0^2}\right\}\right] \quad ✓✓$$

$$= \frac{1}{2}\left[8\sin^{-1}1\right] \quad = 6,283 \; or \quad 2\pi \; units^2 \quad ✓$$

,

(10)

5.1.3

$$\bar{x} = \frac{A_{m-y}}{A}$$

$$A_{m-y} = \int_a^b r dA$$

$$= \int_0^4 x\left[\sqrt{16-x^2} - \frac{1}{2}\sqrt{16-x^2}\right]dx \quad ✓✓✓$$

$$= \frac{1}{2}\int_0^4 x\sqrt{16-x^2}\,dx \quad ✓$$

$$= \frac{1}{2}(-\frac{1}{2})\int_0^4 -2x(16-x^2)^{\frac{1}{2}}\,dx \quad ✓$$

$$= -\frac{1}{4}\left[\frac{\left(16-x^2\right)^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^4 \quad ✓$$

$$= -\frac{1}{6}\left[\left(16-x^2\right)\right]_0^4$$

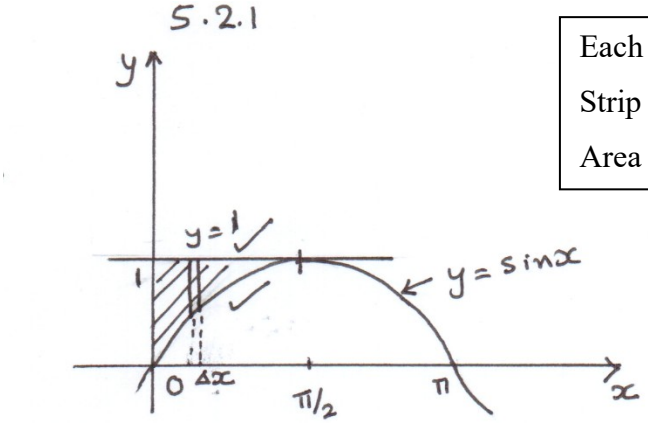
$$= -\frac{1}{6}\left[\left(16-4^2\right)^{\frac{3}{2}} - (16-0^2)^{\frac{3}{2}}\right] \quad ✓$$

$$= \frac{32}{3} units^2 \quad or \quad 10,667 units^2 \quad ✓$$

$$\bar{x} = \frac{32/3}{6,283} = 1,698 \; units \quad ✓✓$$

(10)

5.2 5.2.1




| | |
|---|---|
| Each grapl | ✓✓ |
| Strip | ✓ |
| Area | ✓ |

(4)

5.2.2

$$V_x = \pi\int_a^b y_1^2 - y_2^2 \quad \checkmark$$

$$= \pi\int_0^{\frac{\pi}{2}} 1 - \sin^2 x dx \quad \checkmark\checkmark \quad or \quad \pi\left[x - \left(\frac{x}{2} - \frac{\sin 2x}{4}\right)\right]_0^{\frac{\pi}{2}}$$

$$= \pi\int_0^{\frac{\pi}{2}} \cos^2 x dx \quad \checkmark \qquad = \pi\left[x - \frac{x}{2} + \frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$$

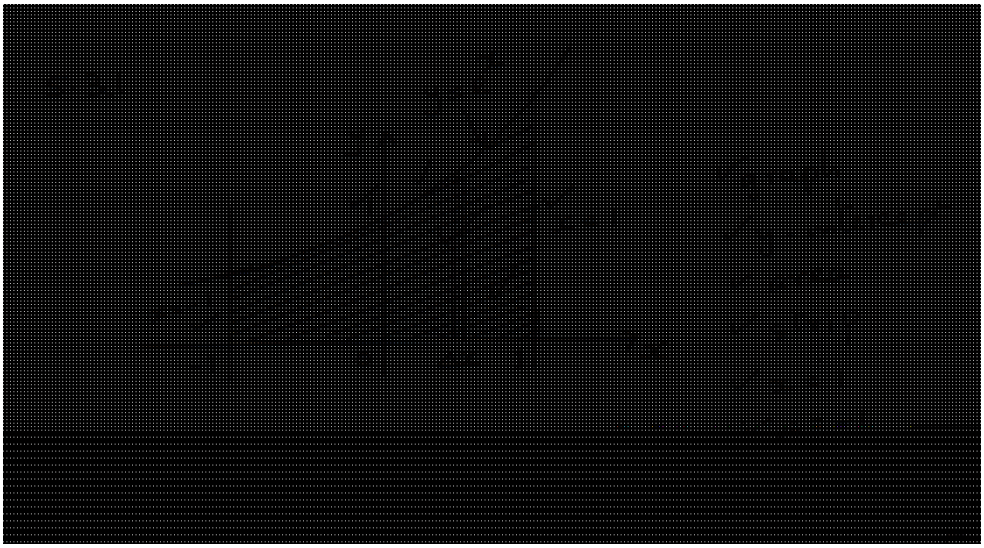
$$= \pi\left[\frac{x}{2} + \frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}} \quad \checkmark \qquad = \pi\left[\frac{x}{2} + \frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$$

$$= \pi\left[\frac{\pi}{4} + \frac{\sin 0}{4} - \{0\}\right] \quad \checkmark\checkmark$$

$$= \frac{\pi^2}{4}\text{units}^3 \quad (\text{or } 2,467) \quad \checkmark$$

(8)

5.3 5.3.1



(6)

5.3.2

$$V_x = \pi\int_a^b y_1^2 - y_2^2 \quad \checkmark$$

$$= \pi\int_{-1}^{1} (e^x)^2 dx \quad \checkmark\checkmark$$

$$= \pi\left[\frac{e^{2x}}{2}\right]_{-1}^{1} \quad \checkmark$$

$$= 3,627\pi \quad \text{or } 11,394\text{units}^3$$

$$= \pi\left[\frac{e^2}{2} - \frac{e^{-2}}{2}\right] \quad \checkmark \qquad or = \frac{\pi}{2}\left[e^2 - e^{-2}\right]$$

$$= 3,627\pi \quad \text{or } 11,394\text{units}^3 \quad \checkmark$$

(6)

5.3.3

$$I_x = \frac{1}{2}\pi\rho\int_a^b y^4\,dx \quad \checkmark\checkmark$$

$$= \frac{1}{2}\pi\rho\int_{-1}^{1}(e^x)^4\,dx \quad \checkmark\checkmark$$

$$= \frac{1}{2}\pi\rho\left[\frac{e^{4x}}{4}\right]_{-1}^{1} \quad \checkmark$$

$$= \frac{1}{2}\pi\rho\left[\frac{e^4}{4} - \frac{e^{-4}}{4}\right] \quad or\ \frac{1}{8}\pi\rho\left(e^4 - e^{-4}\right) \quad \checkmark$$

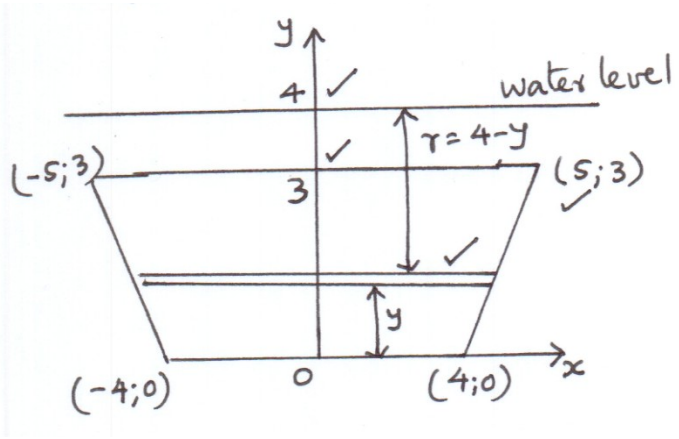
$$= 6,823\pi\rho \quad or\ 21,434\rho \quad \checkmark$$

$$= \frac{6,823\pi m}{3,627\pi} = \frac{21,434m}{11,394} \quad \checkmark\checkmark$$

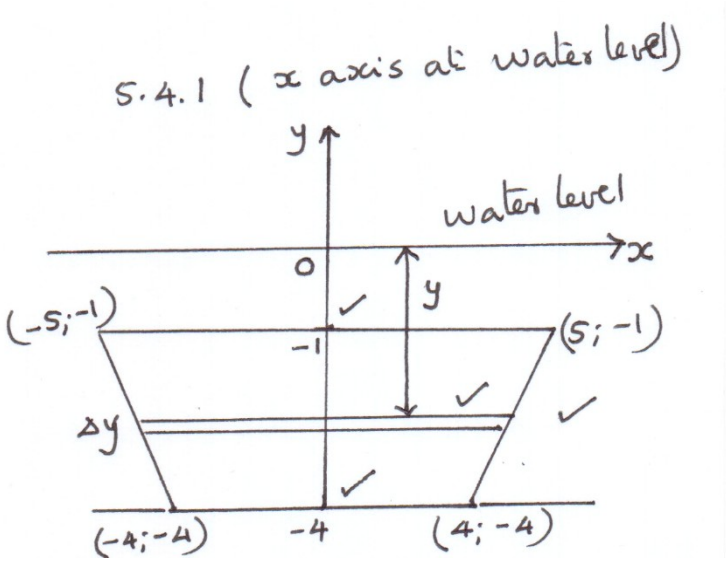
$$= 1,881m \quad \checkmark$$

(10)

5.4 5.4.1




Alternative




(4)

5.4.2 $m=\frac{3-0}{5-4}\ or\ \frac{0-3}{4-5}=3$ ✓

$y-0=3(x-4)\ \ or\ \ y-3=3(x-5)$

$y=3x-12$ ✓ $or\ \ y-3=3x-15$

$or\ \ y=3x-12$

$x=\frac{y+12}{3}$

$=\int_a^b rdA$

$=\int_0^3 (4-y)2\frac{y+12}{3}dy$

$=\frac{2}{3}\int_0^3 (4-y)(y+12)dy$

$=\frac{2}{3}\int_0^3 (-8y-y^2+48)dy$

$=\frac{2}{3}\left[-4y^2-\frac{y^3}{3}+48y\right]_0^3$

$=\frac{2}{3}\left[-4(3)^2-\frac{(3)^3}{3}+48(3)-\{0\}\right]$

$=66m^3$

**Alternative**

With x-axis at the water level

$m=\frac{-1+4}{5-4}\ or\ \frac{-4+1}{4-5}=3$ ✓

$y+4=3(x-4)\ \ or\ \ y+1=3(x-5)$

$y=3x-16$ ✓ $or\ \ y=3x-15-1$

$or\ \ y=3x-16$

$x=\frac{y+16}{3}$

$$\text{First moment of area} = \int_a^b rdA$$

$$= \int_{-4}^{-1} y2\frac{y+16}{3}dy \quad ✓✓$$

$$= \frac{2}{3}\int_{-4}^{-1} y^2 + 16ydy \quad ✓$$

$$= \frac{2}{3}\left[\frac{y^3}{3} + 8y^2\right]_{-4}^{-1} \quad ✓$$

$$= \frac{2}{3}\left[\frac{-1}{3} + 8 - \{\frac{-64}{3} + 8.16\}\right] \quad ✓$$

$$= -66m^3 \qquad (8)$$

5.4.3 Second moment of area

$$= \int_a^b r^2 dA$$

$$= \int_0^3 (4-y)^2 2\frac{y+12}{3}dy \quad ✓✓$$

$$= \frac{2}{3}\int_0^3 (4-y)^2(y+12)dy$$

$$= \frac{2}{3}\int_0^3 (16-8y+y^2)(y+12)dy$$

$$= \frac{2}{3}\int_0^3 (-80y+4y^2+y^3+192)dy \quad ✓$$

$$= \frac{2}{3}\left[-40y^2 + \frac{4}{3}y^3 + \frac{y^4}{4} + 192y\right]_0^3 \quad ✓$$

$$= \frac{2}{3}\left[-40(3)^2 + \frac{4}{3}(3)^3 + \frac{(3)^4}{4} + 192(3) - \{0\}\right] \quad ✓$$

$$= 181,5m^4 \quad ✓$$

$$\bar{y} = \frac{181,5}{66} = 2,75m \quad ✓$$

**Alternative** (8)

Second moment of area

$$= \int_a^b r^2 dA$$

$$= \int_{-4}^{-1} y^2 2\frac{y+16}{3} dy$$

$$= \frac{2}{3}\int_{-4}^{-1} y^3 + 16y^2 dy$$

$$= \frac{2}{3}\left[\frac{y^4}{4} + \frac{16y^3}{3}\right]_{-4}^{-1}$$

$$= \frac{2}{3}\left[\frac{1}{4} - \frac{16}{3} - \left\{\frac{(-4)^4}{4} + \frac{16(-4)^3}{3}\right\}\right]_{-4}^{-1}$$

$$= 181,5m^4$$

$$= \bar{y} = \frac{181,5m^4}{-66m^3} = -2,75m$$

**[80]**

## QUESTION 6

6.1 $x = e^\theta \sin\theta$

$$\frac{dx}{d\theta} = e^\theta \cos\theta + e^\theta \sin\theta \quad \checkmark$$

$$= e^\theta(\cos\theta + \sin\theta)$$

$$\left[\frac{dx}{d\theta}\right]^2 = e^{2\theta}(\cos\theta + \sin\theta)^2 \quad \checkmark$$

$$y = e^\theta \cos\theta \quad \checkmark$$

$$\frac{dy}{d\theta} = -e^\theta \sin\theta + e^\theta \cos\theta \quad \checkmark$$

$$= e^\theta(\cos\theta - \sin\theta)$$

$$\left[\frac{dy}{d\theta}\right]^2 = e^{2\theta}(\cos\theta - \sin\theta)^2 \quad \checkmark$$

$$\left[\frac{dx}{d\theta}\right]^2 + \left[\frac{dy}{d\theta}\right]^2 = e^{2\theta}\left[(\cos\theta + \sin\theta)^2 + (\cos\theta - \sin\theta)^2\right] \quad \checkmark$$

$$= e^{2\theta}\left[\cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta + \sin^2\theta - 2\sin\theta\cos\theta\right] \quad \checkmark$$

$$= e^{2\theta}[2] \quad \checkmark$$

Or using

$a^2+b^2=(a+b)^2-2ab$ $\qquad e^{2\theta}\left[(\cos\theta+\sin\theta)^2+(\cos\theta-\sin\theta)^2\right]=e^{2\theta}[2]$

$$S=\int_{\theta_1}^{\theta_2}\sqrt{\left[\frac{dx}{d\theta}\right]^2+\left[\frac{dy}{d\theta}\right]^2}\,d\theta$$

$$=\int_0^{\frac{\pi}{3}}\sqrt{2e^{2\theta}}\,d\theta \quad \checkmark$$

$$=\sqrt{2}\int_0^{\frac{\pi}{3}}e^{\theta}\,d\theta$$

$$=\sqrt{2}\left[e^{\theta}\right]_0^{\frac{\pi}{3}} \quad \checkmark$$

$$=\sqrt{2}\left[e^{\frac{\pi}{3}}-e^{0}\right] \quad \checkmark$$

$$=2,616 \text{ units} \quad \checkmark$$

(12)

6.2

$$y=\frac{3}{2}x$$

$$\frac{dy}{dx}=\frac{3}{2} \quad \checkmark$$

$$\left[\frac{dy}{dx}\right]^2=\frac{9}{4} \quad \checkmark$$

$$1+\left[\frac{dy}{dx}\right]^2=1+\frac{9}{4}=\frac{13}{4} \quad \checkmark\checkmark$$

$$A_x=2\pi\int_a^b y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \quad \checkmark$$

$$=2\pi\int_2^4\frac{3}{2}x\sqrt{\frac{13}{4}}dx \quad \checkmark\checkmark\checkmark$$

$$=2\pi\frac{3}{2}\frac{\sqrt{13}}{2}\int_2^4 x\,dx \quad \checkmark$$

$$=\frac{3\sqrt{13}\pi}{2}\left[\frac{x^2}{2}\right]_2^4 \quad \checkmark$$

$$=\frac{3\sqrt{13}\pi}{4}\left[4^2-2^2\right] \quad \checkmark$$

$$=9\sqrt{13}\pi \text{ or } 101,945 \text{ units}^2 \quad \checkmark$$

$$x=\frac{2}{3}y$$

$$\frac{dx}{dy}=\frac{2}{3}$$

$$\left[\frac{dx}{dy}\right]^2=\frac{4}{9}$$

$$1+\left[\frac{dx}{dy}\right]^2=1+\frac{4}{9}=\frac{13}{9}$$

$$A_x=2\pi\int_a^b y\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$$

$$=2\pi\int_3^6 y\sqrt{\frac{13}{9}}dy$$

$$=2\pi\frac{\sqrt{13}}{3}\int_3^6 y\,dy$$

$$=\frac{2\sqrt{13}\pi}{3}\left[\frac{y^2}{2}\right]_3^6$$

$$=\frac{\sqrt{13}\pi}{3}\left[6^2-3^2\right]$$

$$=9\sqrt{13}\pi \text{ or } 101,945 \text{ units}^2$$

(12)

**[24]**

**TOTAL: 200**

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